

SANDESH CHAPAGAIN

Backend & Infrastructure Engineer · Real-time media and distributed systems

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PROFILE

Backend and infrastructure engineer working on real-time media. Founder and sole engineer of **Ditch**, a browser-based streaming studio in production that composes, encodes and goes live to YouTube, Twitch, TikTok and Kick from a single tab. I also build small, inspectable systems tools in the open. Second-year Engineering Sciences at Tor Vergata, looking for infrastructure and product engineering roles.

SELECTED WORK

Ditch — Browser-Based Live Streaming Studio

Jan 2026 – Present

Founder & Sole Engineer · Startcup Lazio 2026 · Live in production

- **Studio in the browser:** canvas compositor with scenes, layers and a live preview, holding 30fps even with the tab backgrounded — no install, no hardware encoder.
- **Encode once, push everywhere:** WebCodecs H.264/Opus encoding streamed to a custom relay that fans out to four platforms through ffmpeg; one destination failing never drops the others.
- **Platform:** Fastify API, Socket.io realtime layer, Supabase Postgres with RLS — a TypeScript monorepo of four services on Vercel and Railway.
- **Reliability:** bounded upload buffer with ffmpeg backpressure, single-flight auto-reconnect, and local JWT verification that kept traffic serving through an auth outage.

OPEN SOURCE

- **PulseForge** — durable job orchestration with retries, idempotency and dead letters. [TypeScript](#)
- **SignalScope** — WebRTC observability: quality scoring and anomaly diagnosis. [TypeScript](#)
- **DriftSafe** — static risk analysis for PostgreSQL migrations, SARIF output for CI. [Python](#)
- **CanaryKit** — feature flags with deterministic bucketing and explainable decisions. [TypeScript](#)
- **LocalLens** — citation-first local retrieval, hybrid ranking, no cloud calls. [TypeScript](#)
- **VerityLedger** — double-entry ledger: integer money, hash-chained audit log. [Python](#)
- **ArchScale** — capacity planning for throughput, storage, topology and cost. [TypeScript](#)
- **Roadrash** — pseudo-3D browser racer on Canvas 2D, no game engine. [TypeScript](#)

EXPERIENCE

Network Support Intern · Intrasoft Networking Solutions, Nepal

Jun – Nov 2024

Diagnosed client network faults, monitored with Wireshark and Nagios, configured routers and switches.

Marketing & Growth Manager · Pathik Gyan Niketan, Nepal

Jul 2023 – Jun 2024

Digital presence across three platforms; Meta and Google Ads campaigns with ROI tracking.

Co-creator & Growth Lead · BarcaBuzz

Aug 2022 – Jun 2023

Grew a football community to 100K+ followers organically on a tracked, A/B tested content pipeline.

EDUCATION

B.Sc. Engineering Sciences · Università di Roma Tor Vergata

Dec 2024 – Present

Year 2 of 3 · Mathematics, Physics, Electronics, Computing, Control Systems

TECHNICAL SKILLS

LANGUAGES TypeScript · JavaScript · Python · SQL · Go and C (foundational)

REAL-TIME WebRTC · WebCodecs · RTMP · ffmpeg · Canvas API · Web Workers · WebSockets

BACKEND Node.js · Fastify · Socket.io · PostgreSQL · Supabase (Auth, RLS) · REST · JWT

FRONTEND Next.js (App Router) · React · Zustand · Tailwind CSS

INFRA Docker · Railway · Vercel · Turborepo · GitHub Actions · Git · Linux

SPOKEN Nepali (native) · English (C2) · Hindi (C2) · Italian (B1)